

Summer 1977

Problem 1 (Su77). Prove the following statements about the polynomial ring $\mathbb{F}[x]$, where $\mathbb{F}$ is any field.

1. $\mathbb{F}[x]$ is a vector space over $\mathbb{F}$.
2. The subset of polynomials of degree $\leq n$ is a subspace of dimension $n + 1$ in $\mathbb{F}[x]$.
3. The polynomials $1, x - a, \dots, (x - a)^n$ form a basis of $\mathbb{F}_n[x]$ for any $a \in \mathbb{F}$.

Problem 2 (Su77). Let f be continuous on $\mathbb{C}$ and analytic on $\{z \mid \Im z \neq 0\}$. Prove that f must be analytic on $\mathbb{C}$.

Problem 3 (Su77, Fa81, Sp92, Fa08). Prove that $\alpha = \sqrt{5} + \sqrt{7}$ is algebraic over $\mathbb{Q}$ by writing down a polynomial $f(x) \in \mathbb{Q}[x]$ of degree 4 having α as a zero. Show that $f(x)$ is irreducible over $\mathbb{Q}$.

Problem 4 (Su77, Fa22). Let A be an $r \times r$ matrix of real numbers. Prove that the infinite sum

$$e^A = I + A + \frac{A^2}{2} + \cdots + \frac{A^n}{n!} + \cdots$$

of matrices converges, i.e., for each i, j , the sum of $(i, j)^{th}$ entries converges, and hence that e^A is a well defined matrix.

Problem 5 (Su77, Sp17). Write all values of i^i in the form $a + bi$.

Problem 6 (Su77). Show that $F(k) = \int_0^{\pi/2} \frac{dx}{\sqrt{1 - k \cos^2 x}}$ $0 \leq k < 1$, is an increasing function of k .

Problem 7 (Su77). Let $A : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ be a linear transformation such that $A^{26} = I$. Show that $\mathbb{R}^6 = V_1 \oplus V_2 \oplus V_3$, where V_1, V_2 , and V_3 are two-dimensional invariant subspaces for A .

Problem 8 (Su77, Su80, Sp82, Sp83). Prove that the initial value problem

$$\frac{dx}{dt} = 3x + 85 \cos x \quad x(0) = 77$$

has a solution $x(t)$ defined for all $t \in \mathbb{R}$.

Problem 9 (Su77). Show that every rotation of $\mathbb{R}^3$ has an *axis*; that is, given a 3×3 real matrix A such that $A^T = A^{-1}$ and $\det A > 0$, prove that there is a nonzero vector v such that $Av = v$.

Problem 10 (Su77). Suppose that $f(x)$ is defined on $[-1, 1]$, and that $f'''(x)$ is continuous. Show that the series

$$\sum_{n=1}^{\infty} \left(n \left(f\left(\frac{1}{n}\right) - f\left(-\frac{1}{n}\right) \right) - 2f'(0) \right)$$

converges.

Problem 11 (Su77). Let $f(x, t)$ be a C^1 function such that $\partial f / \partial x = \partial f / \partial t$. Suppose that $f(x, 0) > 0$ for all x . Prove that $f(x, t) > 0$ for all x and t .

Problem 12 (Su77). Let V be the vector space of all polynomials of degree ≤ 10 , and let D be the differentiation operator on V , i.e., $Dp(x) = p'(x)$.

1. Show that $\text{tr } D = 0$.
2. Find all eigenvectors of D and e^D .

Problem 13 (Su77, Sp24). Let f be an analytic function such that

$$f(z) = 1 + 2z + 3z^2 + \cdots \quad \text{for } |z| < 1$$

Define a sequence of real numbers $a_0, a_1, a_2, \dots$ by

$$f(z) = \sum_{n=0}^{\infty} a_n (z + 2)^n$$

Find the radius of convergence of the series

$$\sum_{n=0}^{\infty} a_n z^n$$

Problem 14 (Su77, Sp92). 1. Prove that every finitely generated subgroup of $\mathbb{Q}$, the additive group of rational numbers, is cyclic.

2. Does the same conclusion hold for finitely generated subgroups of $\mathbb{Q}/\mathbb{Z}$, where $\mathbb{Z}$ is the group of integers?

Note: See also ??????.

Problem 15 (Su77). Let $A \subset \mathbb{R}^n$ be compact, $x \in A$; let (x_i) be a sequence in A such that every convergent subsequence of (x_i) converges to x .

1. Prove that the entire sequence (x_i) converges.
2. Give an example to show that if A is not compact, the result in Part 1 is not necessarily true.

Problem 16 (Su77, Fa84, Sp94, Sp96, Sp09, Fa11). Evaluate the integral

$$I(a) = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}$$

where a is real and $a > 1$. Why the formula obtained for $I(a)$ is also valid for certain complex (nonreal) values of a ?

Problem 17 (Su77). In the ring $\mathbb{Z}[x]$ of polynomials in one variable over the integers, show that the ideal $\mathfrak{J}$ generated by 5 and $x^2 + 2$ is a maximal ideal.

Problem 18 (Su77, Sp81). Let $\hat{a}_0 + \hat{a}_1 z + \cdots + \hat{a}_n z^n$ be a polynomial having $\hat{z}$ as a simple zero. Show that there is a continuous function $r : U \rightarrow \mathbb{C}$, where U is a neighborhood of $(\hat{a}_0, \dots, \hat{a}_n)$ in $\mathbb{C}^{n+1}$, such that $r(a_0, \dots, a_n)$ is always a zero of $a_0 + a_1 z + \cdots + a_n z^n$, and $r(\hat{a}_0, \dots, \hat{a}_n) = \hat{z}$.

Problem 19 (Su77). Let p be an odd prime. If the congruence $x^2 \equiv -1 \pmod{p}$ has a solution, show that $p \equiv 1 \pmod{4}$.

Problem 20 (Su77). Determine all solutions to the following infinite system of linear equations in the infinitely many unknowns $x_1, x_2, \dots$:

$$\begin{array}{cccc} x_1 & + & x_3 & + & x_5 & = & 0 \\ x_2 & + & x_4 & + & x_6 & = & 0 \\ x_3 & + & x_5 & + & x_7 & = & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \end{array}$$

How many free parameters are required?